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Defect-induced recombination analysis in 4H-SiC epitaxial layers based on detailed microwave photoconductive decay curve evaluation

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ABSTRACT

Evaluation of defect-related carrier lifetimes in 4H-SiC epitaxial layers is essential for assessing material quality and device performance. The microwave photoconductive decay (μ-PCD) method is widely used for wafer-level lifetime characterization; however, conventional evaluations based on a single lifetime value provide limited insight into defect-induced recombination behavior. In this study, μ-PCD decay curves were analyzed in detail for various defect types in 4H-SiC epitaxial layers to elucidate the impact of defect-induced recombination on excess carrier decay behavior. Polytype inclusions, micropipes, stacking faults, and propagated stacking faults were identified from 1/e lifetime images derived from μ-PCD measurement data. Decay curves acquired at each defect location were analyzed using a multi-component model to separately evaluate the high-intensity (early-time) and low-intensity (late-time) regions. Although all defects appeared as lifetime-reduced regions in 1/e lifetime images, the corresponding μ-PCD decay curves exhibited distinct decay characteristics. Polytype inclusions exhibited both a short-lifetime component consistent with enhanced recombination at the defect interface and a long-lifetime component associated with the defect bulk, whereas planar defects, such as stacking faults, were predominantly governed by short-lifetime interface-related recombination. Furthermore, mapping images that selectively highlighted polytype inclusions were successfully demonstrated by focusing on decay behavior within a specific time window. These results indicate that the temporal characteristics of μ-PCD signals contain physically meaningful information, reflecting both the recombination environment and the cross-sectional structure of epitaxial defects. The proposed analysis framework extends μ-PCD from a conventional lifetime evaluation technique to an advanced, defect-selective characterization method for 4H-SiC epitaxial layers.

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I. INTRODUCTION

Silicon carbide (SiC) is a wide-bandgap semiconductor with excellent physical properties, including high electric-field breakdown strength, high thermal conductivity, and high saturation drift velocity; thus, SiC is a promising material for compact, high-performance, high-efficiency power conversion systems operating at high breakdown voltages.[1–3] SiC-based power devices have been successfully commercialized for applications such as electric vehicles, railways, and industrial power supplies. Nevertheless, continuous effort has been devoted to the development of advanced SiC power devices (including bipolar devices) capable of satisfying the growing demand for higher breakdown voltages and larger currents. The combination of an expanding application space and more stringent technical specifications for SiC devices has consequently increased demand for high-quality SiC wafers and epitaxial layers.

SiC wafers and epitaxial layers inherently contain various crystalline defects—such as threading dislocations, stacking faults (SFs), and basal plane dislocations—originating from bulk crystal growth and epitaxial growth processes.[4,5] These defects can adversely affect device performance and reliability.[6,7] They

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manifest as morphological imperfections and introduce deep energy levels within the bandgap,[8,9] where they function as carrier trapping sites and recombination centers that reduce the carrier lifetime.[10,11] Consequently, crystalline defects increase conduction losses and decrease switching performance and long-term reliability. Therefore, carrier lifetime is regarded as a key metric for evaluating the material quality of SiC.

Numerous studies have investigated the relationship between crystalline defects and carrier lifetime in SiC.[12–17] Lifetime imaging measurements have demonstrated that carrier lifetimes are significantly reduced in regions containing structural defects [e.g., dislocation clusters, polytype inclusions (PIs), and SFs].[12–15] For SFs, complex defects consisting of multiple overlapping SFs exhibit stronger recombination activity than single SFs,[15] while electron-beam-induced current (EBIC) measurements have shown that the carrier lifetime increases for certain SF types.[17] In many previous studies, carrier lifetime evaluation relied on a single lifetime value; however, carrier lifetime reflects the local recombination environment associated with crystalline defects. Given the wide variety of crystalline defect morphologies in SiC and differences in defect-induced electronic states and recombination levels, carrier recombination behavior is expected to vary depending on defect type. Therefore, a detailed analysis of excess carrier decay curves is required to clarify defect-dependent carrier lifetime characteristics; reliance on a single lifetime value alone is insufficient.

Several techniques are commonly used for carrier lifetime measurements in SiC, including microwave photoconductive decay (μ-PCD), time-resolved photoluminescence (TRPL), and free carrier absorption (FCA). The μ-PCD technique enables direct, non-contact, and nondestructive detection of conductivity changes induced by excess carriers with less sensitivity to surface roughness, leading to its widespread adoption for practical wafer-level characterization of SiC materials.[18]

Herein, we validate an advanced inspection approach for SiC epitaxial defects based on a detailed analysis of μ-PCD decay curves. Measurement conditions that enhance defect detection sensitivity and selectivity are identified by incorporating the physical effects of epitaxial defects into a theoretical model of excess carrier decay. The experimental results demonstrate that detailed μ-PCD decay curve analysis is effective for distinguishing among PIs, micropipes (MPs), and SFs based on their characteristic decay behavior, highlighting its potential as an advanced inspection technique for SiC epitaxial layers.

II. EXPERIMENT

The sample used in this study was an n-type 6-in. 4H-SiC epitaxial wafer with a Si-face and a 4° off-cut toward the [11$\bar{2}$0] direction. The epitaxial layer was grown to a thickness of 70 μm by chemical vapor deposition, with a donor concentration of $\sim 1 \times 10^{15}$ cm^{-3}. After epitaxial growth, the defect locations and types in the epitaxial layer were identified using confocal differential interference contrast microscopy and photoluminescence (PL) imaging. PL images were acquired at an excitation wavelength of 313 nm using a charge-coupled device camera with a bandpass filter centered at $\lambda = 450 \pm 50$ nm.

Carrier lifetime measurements were performed at room temperature using the differential μ-PCD method. An yttrium lithium fluoride third-harmonic generation laser ($\lambda = 349$ nm) was used as the excitation source. The laser beam had a diameter of ~1.5 mm, a pulse width of 5 ns, and a photon density of 3×10^{15} cm^{-2} at the sample surface.

The decay behavior of excess carriers was monitored by measuring the temporal variation in the reflected signal induced by irradiation with a 26 GHz microwave. The carrier lifetime was defined as the slope of the microwave reflection decay curve on a semi-logarithmic plot, evaluated from the peak value to 1/e of the peak (hereafter referred to as the 1/e lifetime). The 1/e lifetime images were obtained by scanning the wafer surface with a spatial pitch of 0.1 mm and measuring the carrier lifetime at each position. In the 70 μm thick epitaxial wafer used in this study, some epitaxial defects such as PIs and SFs can extend over a millimeter-scale area and can, therefore, be detected within the μ-PCD measurement area. Because the laser beam diameter was approximately 1.5 mm, the measured decay behavior represents an averaged response within the measurement area. Although a smaller laser beam would improve the spatial resolution, a beam diameter of approximately 1.5 mm was adopted to maintain sufficient μ-PCD signal intensity for reliable decay curve analysis.

μ-PCD decay curves were acquired at defect sites identified from the PL and 1/e lifetime images (where carrier lifetime reduction was observed), and at nearby defect-free reference regions. The temporal origin of each decay curve, $t = 0$, was defined as the time at which the microwave reflectivity reached its maximum intensity after termination of the laser pulse. The decay curves were recorded over a time window of up to ~10 μs following the peak reflectivity. The measured signals were normalized by the peak intensity I_0, and the normalized decay curves $I(t)/I_0$ were analyzed by separating them into an early-time region [$I(t)/I_0 > 10^{-2}$] and a late-time region [$I(t)/I_0 < 10^{-2}$]. Each region was fitted independently to evaluate the defect-dependent decay behavior.

III. μ-PCD DECAY CURVE ANALYSIS AND MODELING

In the μ-PCD method, the temporal decay of the reflected microwave signal is used to detect changes in conductivity induced by excess carriers generated through pulsed optical excitation.[18] An effective carrier lifetime is obtained because multiple recombination processes contribute simultaneously to the observed decay behavior. In general, the effective lifetime τ_{eff} includes contributions from bulk and surface recombination processes and can be expressed as[19]

$$\frac{1}{\tau_{\mathrm{eff}}} = \frac{1}{\tau_{\mathrm{B}}} + \frac{1}{\tau_{\mathrm{s}}}, \tag{1}$$

where τ_{B} and τ_{S} are the bulk and surface recombination components, respectively. The bulk recombination component, τ_{B}, in turn includes Shockley–Read–Hall (SRH), radiative, and Auger recombination processes such that

$$\frac{1}{\tau_{\mathrm{B}}} = \frac{1}{\tau_{\mathrm{SRH}}} + \frac{1}{\tau_{\mathrm{rad}}} + \frac{1}{\tau_{\mathrm{Auger}}}. \tag{2}$$

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Here, τ_{SRH}, τ_{rad}, and τ_{Auger} are the carrier lifetimes associated with SRH, radiative, and Auger recombination processes, respectively. In 4H-SiC, radiative and Auger recombination are only significant for extremely high excess carrier concentrations. Therefore, carrier recombination is dominated by SRH and surface recombination processes under typical operating and measurement conditions. For n-type materials, the SRH recombination lifetime can be approximated as $\tau_{SRH} = \tau_n + \tau_p$ under high-level injection, whereas $\tau_{SRH} = \tau_p$ holds in the low-injection regime. Here, τ_n and τ_p are the electron and hole capture time constants associated with the defect level, respectively.

Because the μ-PCD behavior depends on both the injection level and the recombination environment, a single exponential function cannot always capture the entire decay behavior.[18,20] Instead, the decay can be interpreted as a superposition of components, typically comprising a short-time component that dominates the initial decay and a long-time component that becomes more prominent at later times.[20,21] Moreover, the observed decay curve inherently incorporates contributions from multiple recombination regions when such regions coexist, because the μ-PCD signal reflects conductivity changes induced by excess carriers within the measurement volume;[22] consequently, the measured decay behavior can be expressed as the sum of decay components associated with each recombination region.

Figure 1 illustrates a schematic cross-sectional structure of an epitaxial layer formed on a SiC substrate in which crystalline defects are embedded. The regions contributing to carrier recombination are classified as the near-surface region of the epitaxial layer (epi-surf), the defect-free bulk region of the epitaxial layer (epi-bulk), the interfacial regions between the defect region and the surrounding epitaxial layer (def-inter), and the interior of the defect region (def-bulk). Based on this classification, the μ-PCD decay curve can be approximated as follows:

$$I(t) = A_{epi-surf}\exp\left(-\frac{t}{\tau_{epi-surf}}\right) + A_{epi-bulk}\exp\left(-\frac{t}{\tau_{epi-bulk}}\right) + B_{def-inter}\exp\left(-\frac{t}{\tau_{def-inter}}\right) + B_{def-bulk}\exp\left(-\frac{t}{\tau_{def-bulk}}\right), \quad (3)$$

where $A_{epi\text{-}surf}$, $A_{epi\text{-}bulk}$, $B_{def\text{-}inter}$, and $B_{def\text{-}bulk}$ are the amplitudes of each carrier recombination component and $\tau_{epi\text{-}surf}$, $\tau_{epi\text{-}bulk}$, $\tau_{def\text{-}inter}$, and $\tau_{def\text{-}bulk}$ are the corresponding effective carrier lifetimes. For defect-free substrate regions, only the epi-surf and epi-bulk components contribute to the recombination process ($B_{def-inter} = B_{def-bulk} = 0$) such that Eq. (3) reduces to

$$I_{ref}(t) = A_{epi-surf}\exp\left(-\frac{t}{\tau_{epi-surf}}\right) + A_{epi-bulk}\exp\left(-\frac{t}{\tau_{epi-bulk}}\right). \quad (4)$$

In regions containing defects, the presence of def-inter and def-bulk components depends on cross-sectional defect geometry; finite-volume defects exhibit an additional def-bulk contribution, whereas planar defects with negligible volume do not.

For μ-PCD decay analysis, the measured signal $I(t)$ was normalized by the peak intensity I_0, and the normalized decay curves $I(t)/I_0$ were evaluated using semilogarithmic plots. A change in slope was observed around $I(t)/I_0 \approx 10^{-2}$ (corresponding to a transition in the injection level; see Sec. IV) which could not be properly reproduced when fitting the entire time range with a single exponential function. Therefore, the decay curves were analyzed separately in an early-time region [$I(t)/I_0 > 10^{-2}$] and a late-time region [$I(t)/I_0 < 10^{-2}$].

Fitting was performed in a stepwise manner to enable the recombination characteristics associated with defects to be evaluated independently from the background recombination contributions of the epitaxial layer. First, the decay curves obtained from defect-free reference regions were fitted using Eq. (4) to determine the epitaxial surface- and bulk-related lifetimes. These epitaxial layer-derived components were assumed to be spatially invariant over a lateral range of several millimeters surrounding each defect location. Therefore, for regions containing defects, the epitaxial surface and bulk components extracted from the reference regions were fixed. The decay curves were then fitted using Eq. (3) to determine the additional defect-related interfacial and bulk components.

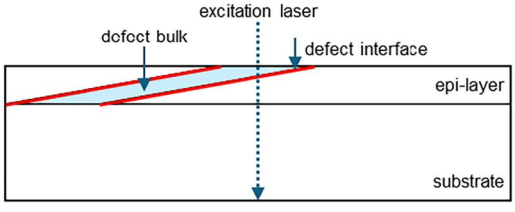


FIG. 1. Schematic cross section of a 4H-SiC epitaxial wafer illustrating the recombination regions considered in the μ-PCD decay curve analysis. The dashed arrow indicates the irradiation direction of the excitation laser.

IV. RESULTS AND DISCUSSION

The μ-PCD measurement results obtained in the vicinity of different defect types were compared using PL images and the corresponding 1/e lifetime images. Figure 2 shows PL and 1/e lifetime images around a PI, MP, SF, and propagated SF (Pro-SF). In the PL images, the PI and MP regions appear dark, whereas SF and Pro-SF are observed as bright regions. For all defect types, a reduction in carrier lifetime is observed at the defect locations in the 1/e lifetime images. The 1/e lifetime predominantly reflects the early-time decay behavior of the μ-PCD signal and does not necessarily represent the full time-dependent behavior of the entire decay curve. The extent of the lifetime reduction varied depending on the defect type, with a particularly pronounced decrease observed for PI compared to the other defects.

Figure 3 shows the μ-PCD decay curves and the corresponding fitting results obtained at a PI and a nearby defect-free reference region. The decay curves were acquired at the approximate positions indicated in Fig. 2(e). In the reference region, distinct slopes were observed in the early- and late-time regions, attributed to changes in the excess carrier concentration during the μ-PCD measurement influencing the dominant carrier recombination

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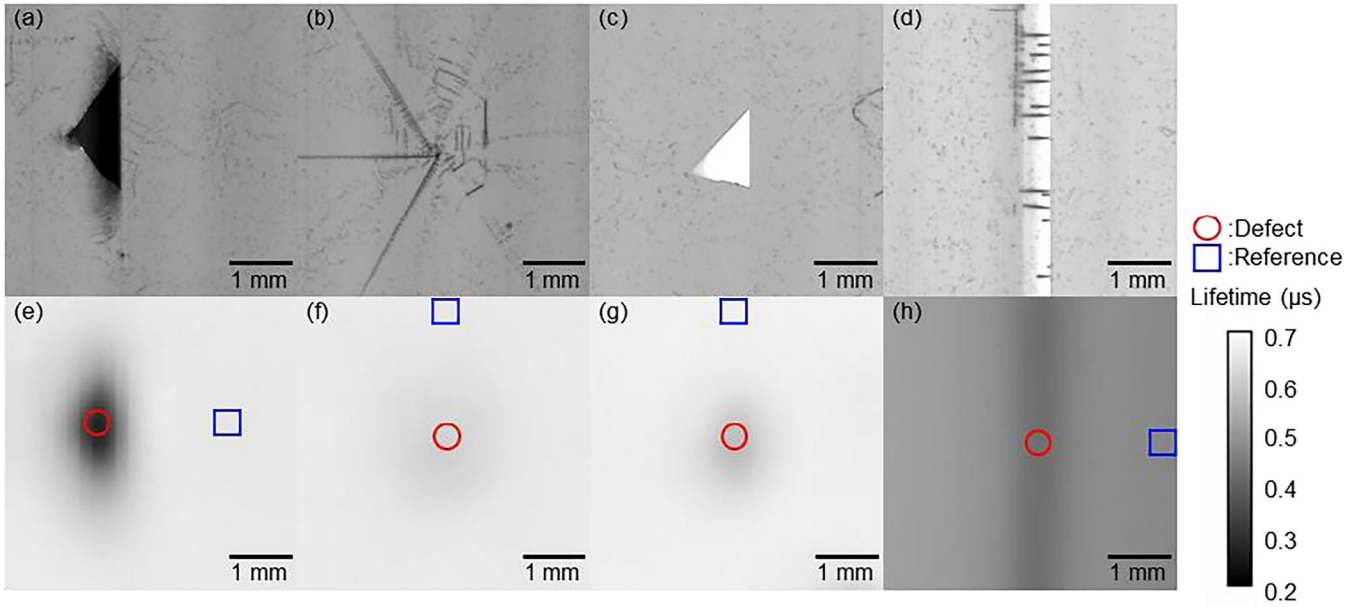


FIG. 2. (a)–(d) PL images and (e)–(h) corresponding 1/e lifetime images obtained by μ-PCD measurements around representative epitaxial defects in 4H-SiC. Panels (a) and (e) correspond to a PI, (b) and (f) to a MP, (c) and (g) to a SF, and (d) and (h) to a Pro-SF. Circles and squares indicate the approximate positions where the μ-PCD decay curves were acquired at defect and nearby defect-free reference regions, respectively.

processes. The excess carrier concentration generated near the epitaxial surface was estimated to be $\sim 1 \times 10^{18}\,\mathrm{cm}^{-3}$ based on the laser photon density of $3 \times 10^{15}\,\mathrm{cm}^{-2}$ and the absorption coefficient $\alpha = 324\,\mathrm{cm}^{-1}$;[23,24] therefore, the decay is considered to start under high-level injection conditions. Previous studies have reported that the μ-PCD signal may exhibit nonlinear behavior under high-level injection conditions.[25,26] Therefore, the carrier lifetimes obtained in this study should be regarded as apparent lifetimes rather than as values directly comparable across different measurement conditions. The high excitation photon density was intentionally selected to obtain a sufficient signal-to-noise ratio for resolving defect-dependent decay behavior over both the early- and late-time regions. Accordingly, the objective of this study is to compare relative differences in decay behavior among different defect types under identical measurement conditions. The carrier concentration corresponding to $I(t)/I_0 \approx 10^{-2}$ was estimated to decrease to $\sim 1 \times 10^{16}\,\mathrm{cm}^{-3}$, approaching the donor concentration of the sample ($1 \times 10^{15}\,\mathrm{cm}^{-3}$), assuming that the μ-PCD signal intensity is approximately proportional to the excess carrier concentration at this signal level. Therefore, in the present study, $I(t)/I_0 \approx 10^{-2}$ was used as a practical boundary corresponding to the transition from high-level toward low-level injection in 4H-SiC, although the exact boundary may vary depending on the doping concentration. In this carrier concentration range, the injection-level dependence of the SRH recombination lifetime may be reflected in the observed decay behavior. Carrier lifetimes in 4H-SiC epitaxial layers have been reported to range from sub-microsecond to several microseconds, depending on factors such as epitaxial layer thickness, doping concentration, defect density, and injection conditions.[21,27] For example, transient FCA and TRPL measurements have determined carrier lifetimes of ~0.8–3 and ~0.1–0.3 μs under high- and low-level injection conditions, respectively.[21] Thus, the epi-bulk components obtained in this study for the

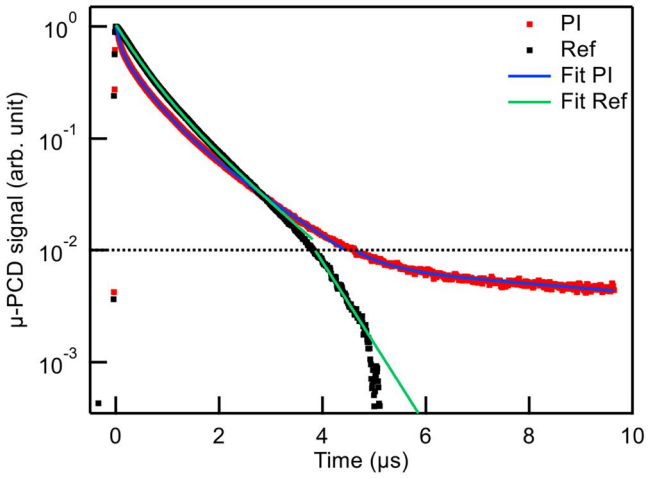


FIG. 3. μ-PCD decay curves and corresponding fitting results obtained at a PI and a nearby defect-free reference region. The decay curves are separately fitted in the early-time and late-time regions separated by the dotted line.

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TABLE I. Effective carrier lifetimes extracted from μ-PCD decay curve fitting for the defect-free reference and PI regions. $\tau_{epi\text{-}surf}$, $\tau_{epi\text{-}bulk}$, $\tau_{def\text{-}inter}$, and $\tau_{def\text{-}bulk}$ are the effective lifetimes associated with the epitaxial surface, epitaxial bulk, defect interface, and defect bulk components, respectively. All lifetime values are given in microseconds (μs). "–" indicates that the parameter was not assessed for that measurement; "fix" indicates that the parameter was fixed during the fitting procedure.

	Early-time region				Late-time region	
Defect type	$\tau_{epi\text{-}surf}$	$\tau_{epi\text{-}bulk}$	$\tau_{def\text{-}inter}$	$\tau_{def\text{-}bulk}$	$\tau_{epi\text{-}bulk}$	$\tau_{def\text{-}bulk}$
Ref	0.46	1.09	…	…	0.60	…
PI	0.46 fix	1.09 fix	0.08	1.65	0.60 fix	11.55

reference region (see Table I) lie within the typical lifetime for 4H-SiC epitaxial layers. The present study focused on a 70 μm-thick epitaxial wafer with a donor concentration of $\sim 1 \times 10^{15}$ cm^{-3}, which is representative of high-voltage device applications. This wafer was selected because its relatively slow decay was suitable for detailed evaluation of defect-dependent differences in the μ-PCD decay curves and for resolving multiple decay components. More commonly used epitaxial wafers typically have thinner epitaxial layers ($\sim$10 μm) and higher doping concentrations ($\sim 1 \times 10^{16}$ cm^{-3}). In such wafers, the relative contribution of surface recombination is expected to increase, resulting in a shorter effective carrier lifetime. Consequently, the decay characteristics may differ from those reported here, and the evaluation time range and analysis conditions may require optimization depending on wafer specifications.

The decay curve measured at the PI is significantly steeper in the early-time region [$I(t)/I_0 > 10^{-2}$] than that measured in the reference region. As time progresses, the decay rate at the PI decreases and a pronounced long-tail component is clearly visible. To quantitatively evaluate this behavior, the decay curve acquired in the reference region was fitted using Eq. (4), and the effective carrier lifetimes associated with the epi-surf and epi-bulk components were extracted. Next, the decay curve obtained at the PI was fitted using Eq. (3), while fixing the epi-surf and epi-bulk components determined from the reference region. Through this fitting procedure, additional defect-related components corresponding to the def-inter and def-bulk components were determined. Table I summarizes the effective carrier lifetimes obtained from the fitting for both the PI and reference regions. In the early-time region, two defect-related components were identified at the PI: a short-lifetime def-inter component ($\tau_{def\text{-}inter} = 0.08\,\mu$s) consistent with enhanced recombination at the defect interface, and a longer-lifetime def-bulk component ($\tau_{def\text{-}bulk} = 1.65\,\mu$s) originating from the defect bulk. In the late-time region, a long-lifetime component >10 μs was obtained at the PI, and the decay behavior was significantly more gradual than that observed in the reference region.

The multiple decay components at the PI can be interpreted based on the cross-sectional defect structure shown in Fig. 4(a); a PI consists of a finite-volume 3C-SiC region embedded within the 4H-SiC epitaxial layer and is characterized by the presence of a 3C/4H-SiC heterointerface. Consequently, the decay of excess carriers involves multiple recombination pathways, including

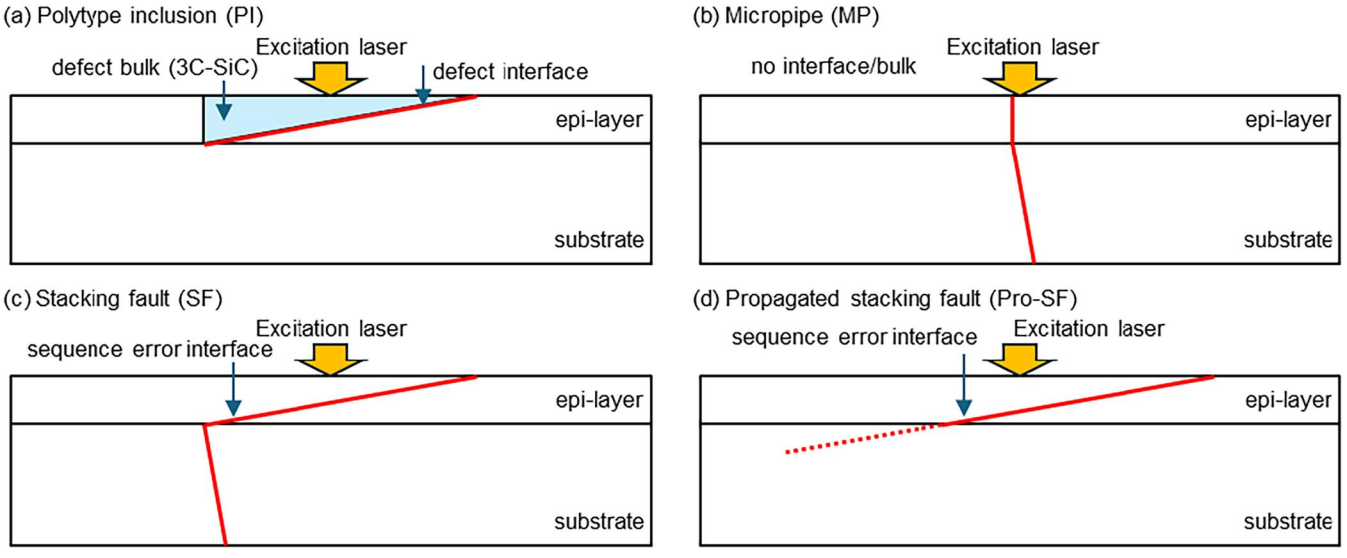


FIG. 4. Schematic cross-sectional illustrations of representative epitaxial defects in 4H-SiC: (a) polytype inclusion (PI), (b) micropipe (MP), (c) stacking fault (SF), and (d) propagated stacking fault (Pro-SF). Yellow arrows indicate the excitation laser.

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recombination near the epitaxial surface, within the epitaxial bulk, at the defect interface, and inside the defect bulk. The short-lifetime def-inter component is consistent with enhanced recombination at the 3C/4H-SiC interface due to a high density of recombination centers or an increased recombination rate associated with the structural and electronic discontinuities induced by defect formation.

Furthermore, pronounced tail components and long carrier lifetimes have been reported for μ-PCD decay curves measured in 3C-SiC, with hole-capture-limited recombination mechanisms proposed.[28–31] Therefore, the observation that the def-bulk component at the PI exhibits a longer lifetime than that of the reference region in both the early- and late-time regions may reflect the intrinsic recombination characteristics of 3C-SiC. In the late-time region—where the recombination process is dominated by the minority carrier lifetime—the long-lifetime def-bulk component observed at the PI suggests the involvement of a hole-related recombination mechanism that differs from that in 4H-SiC. However, the different optical absorption coefficients of 3C-SiC and 4H-SiC make a direct quantitative comparison of carrier lifetime values between the PI and reference regions difficult. Because the absorption coefficient of 3C-SiC is higher than that of 4H-SiC, the excess carrier concentration generated within the PI region may differ from that in the surrounding 4H-SiC. Therefore, contributions from radiative and Auger recombination cannot be completely excluded. In addition, the local PI thickness gradually decreases as the measurement position shifts along the off-cut direction within an individual PI, whereas the lateral area of the PI included within the measurement spot increases. Because the present measurements were performed using a laser beam diameter of approximately 1.5 mm, the observed decay behavior represents an averaged response reflecting the local thickness and lateral extent of the PI within the measurement area. Further investigation using a smaller excitation beam may provide additional insight into the relationship between PI geometry and decay characteristics.

The results obtained by applying the same decay curve analysis to MP, SF, and Pro-SF are shown in Fig. 5 and summarized in Table II. The decay curves were acquired at the approximate positions indicated in Figs. 2(f)–2(h). In the early-time region, no independent defect-related components were extracted for MP, and the decay behavior was similar to that observed in the reference region. Contrastingly, a def-inter component with a shorter lifetime than that of the reference region was extracted for both SF and Pro-SF; however, no independent def-bulk component was obtained for these defects. In the late-time region, extraction of multiple decay components was not feasible for MP, SF, or Pro-SF. Therefore, the decay curves were analyzed using a single exponential function; only Pro-SF exhibited a more gradual decay compared with the reference region, corresponding to a longer effective carrier lifetime.

These differences observed between defect types can be systematically explained by their cross-sectional geometries, as illustrated in Figs. 4(b)–4(d). MPs are extended defects that propagate along the c axis direction. Consequently, the effective defect interface and defect bulk within the excitation and detection volume of the μ-PCD measurement are limited; therefore, the decay behavior does not diverge significantly from that of the reference region. SFs and Pro-SFs are planar defects with atomic-scale thicknesses originating from errors in the stacking sequence. Consequently,

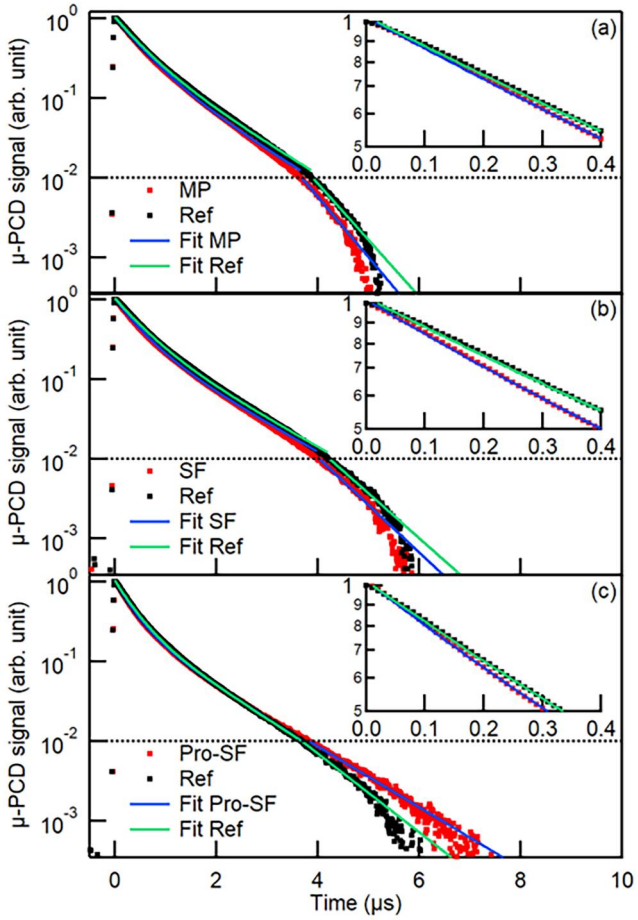


FIG. 5. μ-PCD decay curves and corresponding fitting results obtained at (a) MP, (b) SF, (c) Pro-SF, and their respective defect-free reference regions. The decay curves are separated into early-time and late-time regions by the dotted lines. Insets show magnified views of the initial decay behavior.

TABLE II. Summary of effective carrier lifetimes extracted from μ-PCD decay curve analysis for different defect types. The extracted lifetimes are listed for MP, SF, Pro-SF, and the corresponding defect-free reference region. All lifetime values are given in microseconds (μs). "–" indicates that the parameter was not assessed for that measurement; "fix" indicates that the parameter was fixed during the fitting procedure; "N.D." indicates that the value could not be determined.

Defect type	Early-time region				Late-time region
	$\tau_{\text{epi-surf}}$	$\tau_{\text{epi-bulk}}$	$\tau_{\text{def-inter}}$	$\tau_{\text{def-bulk}}$	$\tau_{\text{epi-bulk/def-bulk}}$
Ref	0.34–0.47	1.06–1.22	…	…	0.61–0.88
MP	0.47 fix	1.11 fix	N.D.	N.D.	0.57
SF	0.47 fix	1.22 fix	0.30	N.D.	0.73
Pro-SF	0.34 fix	1.06 fix	0.28	N.D.	1.14

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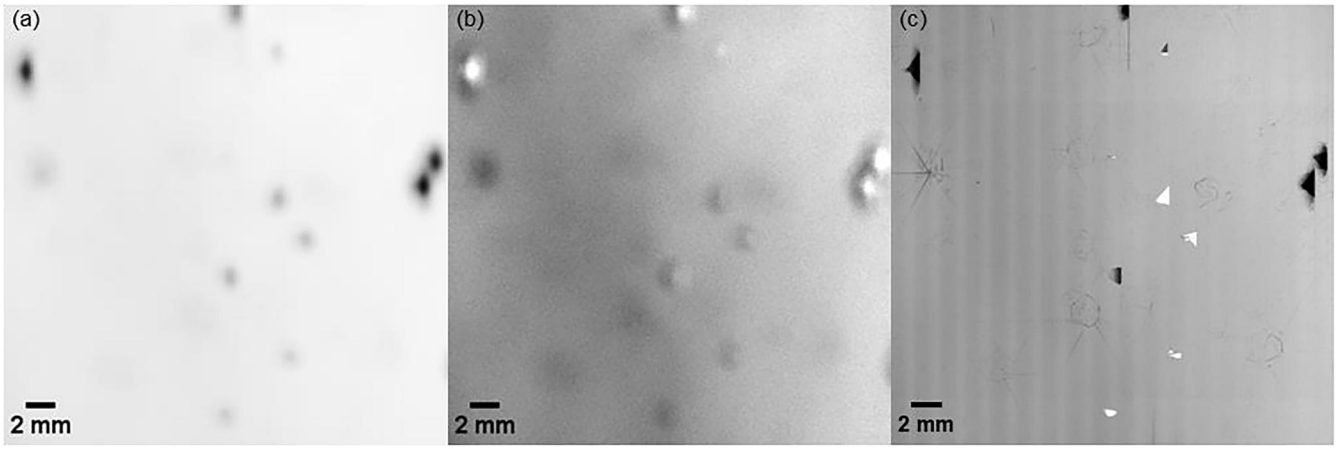


FIG. 6. Wide-area images of (a) 1/e lifetime, (b) τ_2, and (c) PL. In the 1/e lifetime image, defects appear as regions of reduced carrier lifetime, whereas in the τ_2 image, PIs are selectively highlighted with enhanced contrast.

these defects have little effective bulk, and no independent def-bulk component is observed. Instead, only a def-inter component associated with enhanced recombination at the defect interface appears in the early-time region. In the late-time region, MPs, SFs, and Pro-SFs lack an effective bulk defect, and the decay behavior can be described by a single epitaxial bulk-related component. Thus, the presence or absence of decay components extracted from fitting the μ-PCD decay curves, as well as their dependence on defect type, can be systematically understood in terms of differences in defect cross-sectional geometries, specifically the existence of defect interfaces and defect volumes.

Regarding carrier lifetime differences relative to the reference region, the short-lifetime def-inter components observed in the early-time region reflect enhanced recombination at the defect interface (consistent with the behavior discussed for PI). Previous PL studies have revealed characteristic SF-related luminescence and its rapid decay. In particular, TRPL measurements have shown that luminescence from single Shockley stacking faults (1SSFs) decays faster than the band-edge luminescence, indicating enhanced carrier recombination in 1SSFs.[32,33] In 4H-SiC bipolar diodes, SF formation and expansion have been associated with increased forward voltage. Carrier recombination around 1SSFs has also been reported to suppress conductivity modulation and limit the forward current.[34,35] The differences observed in the late-time region may be related to the fact that SFs can function as quantum wells in the band structure.[36–39] Local measurements using EBIC and cathodoluminescence techniques have demonstrated that SFs function as quantum wells and modify the recombination behavior of minority carriers.[15,17] In particular, the minority carrier lifetime has been reported to either decrease or increase depending on the relative position of the quantum-well levels with respect to the Fermi level.[17] The μ-PCD decay curve reflects the temporal evolution of excess carriers; therefore, the lifetime extracted from μ-PCD measurements (for all defect types studied here) not only includes contributions from recombination processes but also from carrier diffusion and transport. When quantum-well levels associated with SFs are located near the Fermi level, they are not necessarily highly occupied under equilibrium conditions. These states can accommodate excess carriers, influencing their spatial distribution and transport near the defect. Moreover, shallow levels or quantum-well structures near the Fermi level do not necessarily serve as strong recombination centers. Under low-level injection conditions, recombination may become limited by hole capture, thereby reducing the contribution of these states to the recombination process. In such cases, the quantum well may function not as a dominant recombination center but as a structure that temporarily localizes carriers and redistributes them spatially.

To further emphasize differences in decay behavior, an additional parameter (τ_2) was calculated from the slope of the decay curve in the time range $t = 1.0$–$2.0\,\mu$s. The time window intentionally corresponded to a regime in which the contribution of defect-bulk-related recombination became non-negligible relative to that of the epitaxial bulk, while the surface-related recombination component had largely decayed. It should be noted that the criterion of $I(t)/I_0 < 10^{-2}$ was introduced for separating the decay curves into early- and late-time regions for curve fitting, whereas τ_2 was defined as an imaging parameter to emphasize differences in defect-dependent decay behavior. Consequently, τ_2 serves as an example parameter that selectively reflects decay components associated with the internal structure of defects. In the present wafer, τ_2 was selected to highlight the long-lifetime component associated with PIs. Because the optimal evaluation time range depends on wafer specifications and carrier lifetime characteristics, τ_2 is not intended as a universal parameter for characterizing slow decay components. The applicability of a single criterion for slow decay components to different wafer specifications requires further investigation. Figure 6 shows wide-area images of the 1/e lifetime, τ_2, and corresponding PL. In the 1/e lifetime image, all defects appear as dark features, reflecting a reduction in carrier lifetime at the defect locations. On the other hand, only the PI appears as a

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bright feature in the τ_2 image. Although some SFs also exhibit weak bright contrast, the contrast enhancement is significantly smaller than that of the PI, possibly reflecting differences in the SF structure and morphology. Because τ_2 is evaluated within a fixed time window, contributions from multiple decay components may still be included. Nevertheless, the pronounced contrast enhancement of the PI suggests that the contribution of the def-bulk component is relatively more significant within this time range than for the other defects.

Overall, these results indicate that, while the 1/e lifetime image is effective for identifying the locations of defects, the decay components extracted through multicomponent analysis of μ-PCD decay curves reflect the characteristic cross-sectional structure of each defect type. Furthermore, by selecting time regions corresponding to specific decay components associated with defects, it becomes possible to selectively enhance the contrast of certain defect types. These findings indicate that detailed analysis of μ-PCD decay behavior may assist in distinguishing among different defect types based on their characteristic decay components.

V. CONCLUSION

Carrier lifetime characteristics associated with various epitaxial defects in 4H-SiC were investigated through detailed analysis of μ-PCD decay behavior. PIs, MPs, SFs, and Pro-SFs were all observed as lifetime-reduced regions in the 1/e lifetime images. However, the μ-PCD decay curves exhibited defect-type-dependent behaviors. By separating the decay curves into early- and late-time regions, recombination components associated with defect interfaces and bulks were selectively extracted. The presence or absence of these components systematically correlated with the cross-sectional geometry of each defect type. In particular, PI exhibited a long-lifetime component associated with the defect bulk, whereas planar defects such as SF and Pro-SF exhibited only short-lifetime components consistent with enhanced recombination at the defect interface. Furthermore, certain defect types were selectively emphasized in lifetime images by focusing on specific time ranges of decay behavior. This result demonstrates that the temporal hierarchy of recombination processes governed by defect geometry can be accessed through appropriate selection of the analysis time window.

Overall, these results demonstrate that the temporal characteristics of μ-PCD decay curves provide physically meaningful insight into both recombination processes and structural features of epitaxial defects, providing information that may assist in the detection and differentiation of defect types based on their characteristic decay behavior. The proposed analysis framework extends the role of μ-PCD beyond conventional lifetime evaluation, establishing it as an advanced inspection technique capable of defect-selective characterization in SiC epitaxial layers.

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AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

Takumi Wakabayashi: Conceptualization (equal); Data curation (lead); Formal analysis (lead); Methodology (lead); Validation (equal); Visualization (lead); Writing – original draft (lead); Writing – review & editing (equal). **Kazushi Hayashi:** Conceptualization (equal); Data curation (supporting); Formal analysis (supporting); Investigation (lead); Methodology (supporting); Resources (equal); Supervision (supporting); Validation (equal); Writing – review & editing (equal). **Hideo Fujii:** Conceptualization (equal); Project administration (equal); Writing – review & editing (supporting). **Naoki Okano:** Conceptualization (equal); Funding acquisition (lead); Project administration (equal); Resources (supporting); Writing – review & editing (supporting). **Junji Senzaki:** Conceptualization (equal); Data curation (supporting); Formal analysis (supporting); Funding acquisition (lead); Investigation (supporting); Methodology (supporting); Project administration (equal); Resources (equal); Supervision (lead); Writing – review & editing (equal).

DATA AVAILABILITY

The data that support the findings of this study are available within the article.

REFERENCES

[1]H. Matsunami and T. Kimoto, Mater. Sci. Eng. R **20**, 125 (1997).
[2]M. Bakowski, IEEJ Trans. Ind. Appl. **126**, 391 (2006).
[3]T. Kimoto, Jpn. J. Appl. Phys. **54**, 040103 (2015).
[4]International Electrotechnical Commission, IEC 63068-1:2019, 2019.
[5]Japan Electronics and Information Technology Industries Association, JEITA EDR-4712/100, 2016.
[6]H. Fujiwara, T. Kimoto, T. Tojo, and H. Matsunami, Appl. Phys. Lett. **87**, 051912 (2005).
[7]L. Lai, Y. Cui, Y. Zhong, K. Y. Cheong, H. Linewih, X. Xu, and J. Han, J. Appl. Phys. **137**, 060701 (2025).
[8]A. Galeckas, A. Hallén, S. Majdi, J. Linnros, and P. Pirouz, Phys. Rev. B **74**, 233203 (2006).
[9]M. E. Bathen, C. T.-K. Lew, J. Woerle, C. Dorfer, U. Grossner, S. Castelletto, and B. C. Johnson, J. Appl. Phys. **131**, 140903 (2022).
[10]P. B. Klein, B. V. Shanabrook, S. W. Huh, A. Y. Polyakov, M. Skowronski, J. J. Sumakeris, and M. J. O'Loughlin, Appl. Phys. Lett. **88**, 052110 (2006).
[11]W. Shockley and W. T. Read, Jr., Phys. Rev. **87**, 835 (1952).
[12]J. Hassan and J. P. Bergman, J. Appl. Phys. **105**, 123518 (2009).
[13]B. Kallinger, M. Rommel, L. Lilja, J. Hassan, I. Booker, E. Janzén, and P. Bergman, Mater. Sci. Forum **778–780**, 301–304 (2014).
[14]Y. Cui, J. Li, K. Zhou, X. Zhang, and G. Sun, Diamond Relat. Mater. **92**, 25–31 (2019).
[15]P. Hou, P. Wang, Y. Li, W. Zhong, Y. Han, J. Wang, L. Yu, Z. Li, and R. Jin, J. Cryst. Growth **650**, 128005 (2025).
[16]S. I. Maximenko, J. A. Freitas, Jr., P. B. Klein, A. Shrivastava, and T. S. Sudarshan, Appl. Phys. Lett. **94**, 092101 (2009).
[17]B. Chen, H. Matsuhata, T. Sekiguchi, A. Kinoshita, K. Ichinoseki, and H. Okumura, Appl. Phys. Lett. **100**, 132108 (2012).

10 October 2026 09:59:56

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[18]T. Asada, Y. Ichikawa, and M. Kato, J. Vis. Exp. **146**, e59007 (2019).
[19]D. K. Schroder, *Semiconductor Material and Device Characterization*, 3rd ed. (Wiley, Hoboken, NJ, 2006).
[20]R. J. Kumar, J. M. Borrego, R. J. Gutmann, J. R. Jenny, D. P. Malta, H. M. Hobgood, and C. H. Carter, Jr., J. Appl. Phys. **102**, 013704 (2007).
[21]P. B. Klein, J. Appl. Phys. **103**, 033702 (2008).
[22]M. Kunst and G. Beck, J. Appl. Phys. **60**, 3558–3566 (1986).
[23]S. G. Sridhara, T. J. Eperjesi, R. P. Devaty, and W. J. Choyke, Mater. Sci. Eng. B **61–62**, 229–233 (1999).
[24]T. Hayashi, K. Asano, J. Suda, and T. Kimoto, J. Appl. Phys. **109**, 014505 (2011).
[25]K. Lauer, A. Laades, H. Übensee, H. Metzner, and A. Lawerenz, J. Appl. Phys. **104**, 104503 (2008).
[26]M. Kato, Y. Mori, and M. Ichimura, Jpn. J. Appl. Phys. **54**, 04DP14 (2015).
[27]L. Lilja, I. D. Booker, J. Hassan, E. Janzén, and J. P. Bergman, J. Cryst. Growth **381**, 43–50 (2013).
[28]M. Ichimura, H. Tajiri, Y. Morita, N. Yamada, and A. Usami, Appl. Phys. Lett. **70**, 1745–1747 (1997).
[29]M. Ichimura, N. Yamada, H. Tajiri, and E. Arai, J. Appl. Phys. **84**, 2727–2731 (1998).
[30]J. W. Sun, I. G. Ivanov, R. Liljedahl, R. Yakimova, and M. Syväjärvi, Appl. Phys. Lett. **100**, 252101 (2012).
[31]M. Ichimura, Solid State Electron. **50**, 1761–1766 (2006).
[32]S. G. Sridhara, F. H. C. Carlsson, J. P. Bergman, and E. Janzén, Appl. Phys. Lett. **79**, 3944–3946 (2001).
[33]M. Kato, S. Katahira, Y. Ichikawa, S. Harada, and T. Kimoto, J. Appl. Phys. **124**, 095702 (2018).
[34]J. D. Caldwell, R. E. Stahlbush, E. A. Imhoff, K. D. Hobart, M. J. Tadjer, Q. Zhang, and A. Agarwal, J. Appl. Phys. **106**, 044504 (2009).
[35]S. Asada, K. Murata, and H. Tsuchida, J. Appl. Phys. **136**, 175702 (2024).
[36]H. Iwata, U. Lindefelt, S. Öberg, and P. R. Briddon, Phys. Rev. B **65**, 033203 (2001).
[37]M. S. Miao and W. R. L. Lambrecht, Phys. Rev. B **68**, 155320 (2003).
[38]U. Lindefelt, H. Iwata, S. Öberg, and P. R. Briddon, Phys. Rev. B **67**, 155204 (2003).
[39]M.-S. Miao and W. R. L. Lambrecht, Phys. Rev. B **85**, 205318 (2012).

10 October 2026 09:59:56

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